# Title line 1
# Title line 2


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About our group

## Introduction

- **Images classification:** 4 datasets about remote sensing, medical imaging, handwritten digits, and fashion.
- **A hybrid protocol:** A simple classical neural network + Hamiltonian time evolution of a spin chain.
- **Non-equilibrium dynamics:** Few qubits significantly enhance the classification accuracy.

**Result subblock**

Identical initial states + Hamiltonian time evolution under different parameters = distinguishable final states.

## Methods

**Datasets**

**Result subblock**

$$H = J \sum_{j=1}^{N_q-1} \sigma_j^x \sigma_{j+1}^x + \sum_{j=1}^{N_q} h_j \sigma_j^z,$$

**Result subblock**

$$O_\alpha = \frac{1}{N_S} \sum_{k=1}^{N_S} |\Psi_k^{(\alpha)}(t)\rangle \langle \Psi_k^{(\alpha)}(t)|, \alpha = 1, 2, \ldots, N_c.$$

**Result subblock**

1. **Exact expectation:** $\langle O_\alpha \rangle = \langle \Psi | O_\alpha | \Psi \rangle$.
2. **Shadow tomography:** $\langle O_\alpha \rangle = \frac{1}{M} \sum_{m=1}^{M} \mathrm{Tr}\left(\rho^{(m)} O_\alpha\right)$.

**Result subblock**

$$C = \frac{1}{N_{trn}} \sum_{\{\mathbf{x_1}, \mathbf{x_2}\}} \left| |\langle \Psi_1(t) | \Psi_2(t) \rangle|^2 - \delta y_1, y_2 \right|.$$

## Model

## Result block 1

**Result subblock**

**Result subblock**

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## Result block 2

**Result subblock**

**Result subblock**

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